

FUNÇÕES

Domínio:

↳ Conjunto de valores que podem ser colocados como entrada $f(x)$.

Contradomínio:

$f: A \rightarrow B$
domínio $\rightarrow$ Contradomínio

Imagem:

$Im(f) \subset CD$

↳ Valores que $f(x)$ assume.

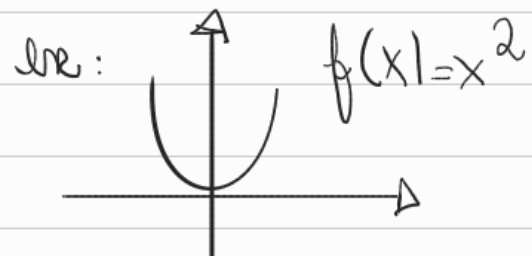
Obs: Intervalos

$] [/ () \rightarrow$ Aberto

$[] \rightarrow$ Fechado.

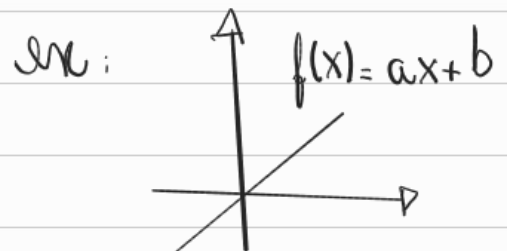
Função Par:

- $f(x) = f(-x)$
- Simetria em relação ao y.
- $\cos(x) / \sec(x)$



Função Ímpar

- $f(x) = -f(-x)$
- Simetria em relação à origem.
- $\sin(x) / \csc(x) / \tan(x)$



Função Injetora:

- $x_1 \neq x_2$ em $A \rightarrow f(x_1) \neq f(x_2)$ em B .

○ Função Sobrejetora:
 $\text{Im}(f) = \text{CD}$

○ Função Bijetora:
 Injetora e Sobrejetora

○ Função Inversa:

$$f(x) = f(x)^{-1}$$

$$f^{-1}: B \rightarrow A$$

$$* f \circ f^{-1}(x) = x$$

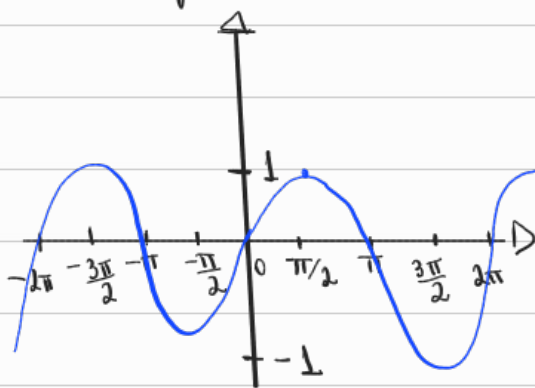
$$* f^{-1} \circ f(x) = x$$

$$* P(a, b) \in \text{gr}(f) \iff P'(b, a) \in \text{gr}(f^{-1})$$

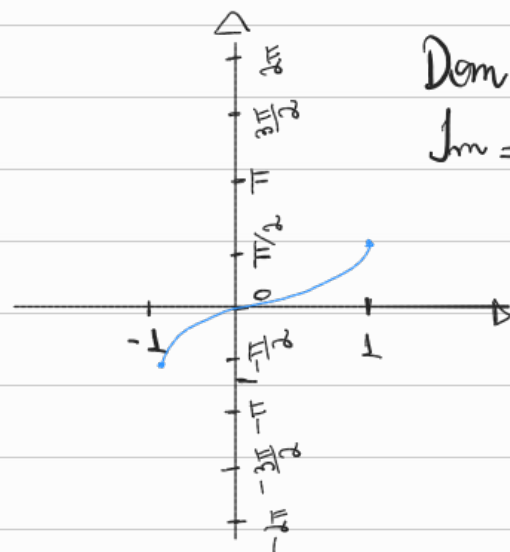
! A função inversa é
 bijetora

○ Funções Trigonométricas:

$$f(x) = \sin x$$



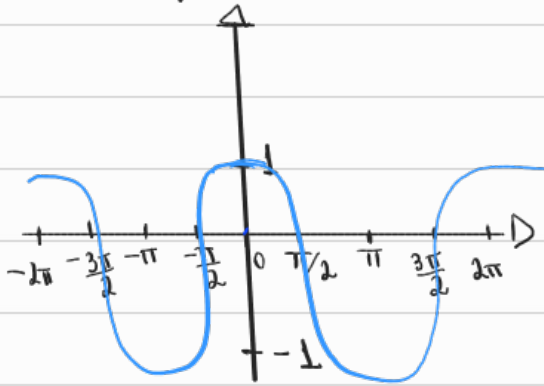
$$f(x) = \arcsin(x)$$



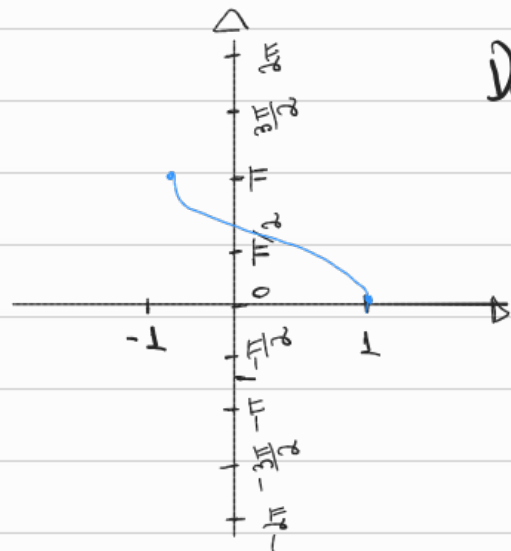
$$\text{Dom} = [-\pi/2, \pi/2]$$

$$\text{Im} =$$

$$f(x) = \cos x$$

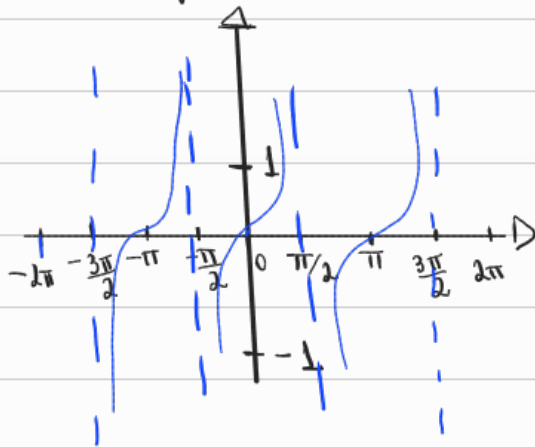


$$f(x) = \arccos(x)$$

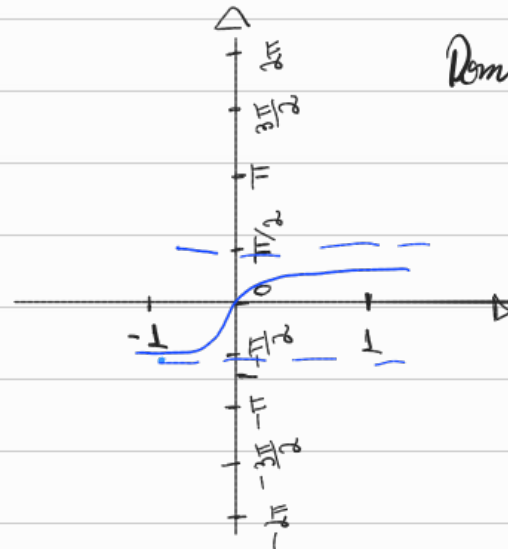


$$\text{Dom} [0, \pi]$$

$$f(x) = \tan(x)$$

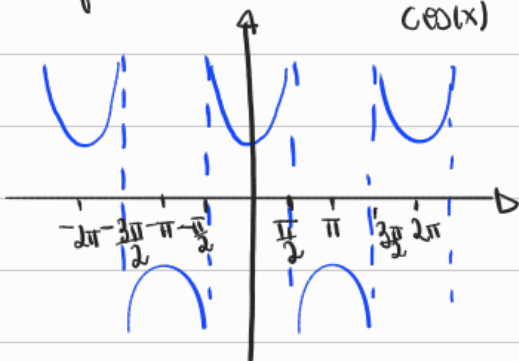


$$f(x) = \arctan(x)$$

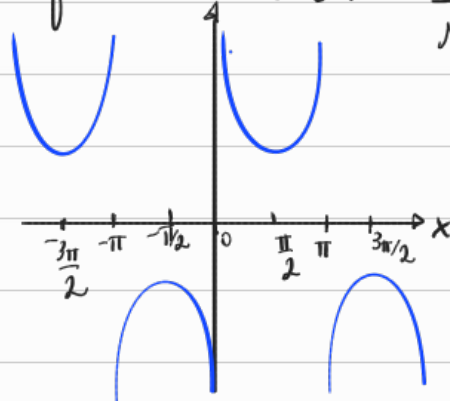


$$\text{Dom} = \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$f(x) = \sec(x) = \frac{1}{\cos(x)}$$



$$f(x) = \csc(x) = \frac{1}{\sin(x)}$$



$$\text{Dom} = \{x \in \mathbb{R} \mid x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z}\}$$

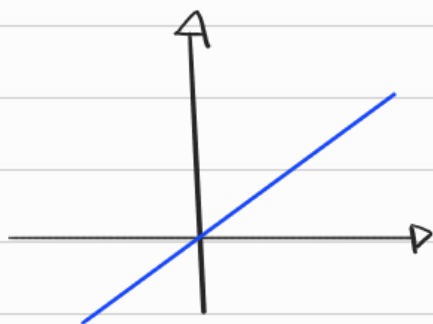
$$\text{Im} =]-\infty, -1] \cup [1, +\infty[$$

$$\text{Dom} = \{x \in \mathbb{R} \mid x \neq k\pi, k \in \mathbb{Z}\}$$

$$\text{Im} =]-\infty, -1] \cup [1, +\infty[.$$

↳ Tipos e Exemplos de Funções:

o Linear:



$$f(x) = x$$

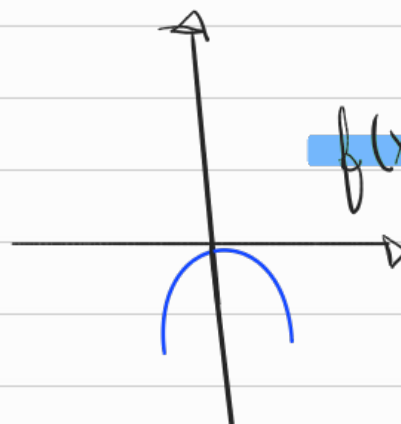


$$f(x) = -x$$

o Quadrática:

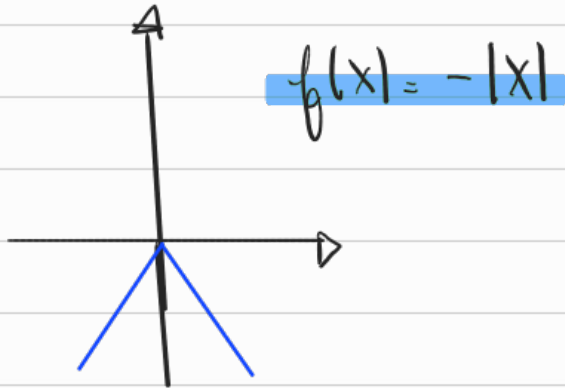
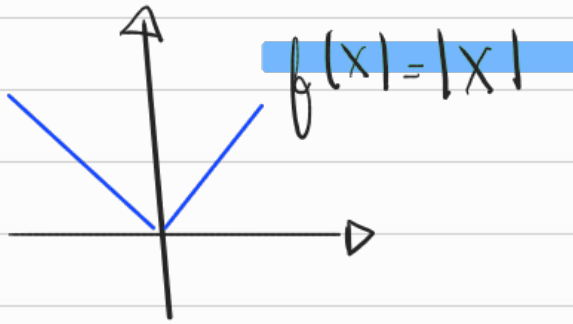


$$f(x) = x^2$$

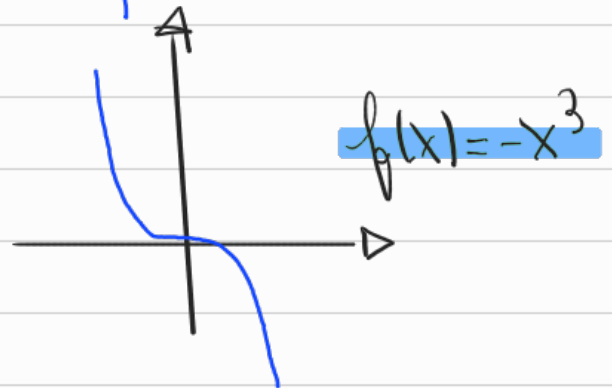
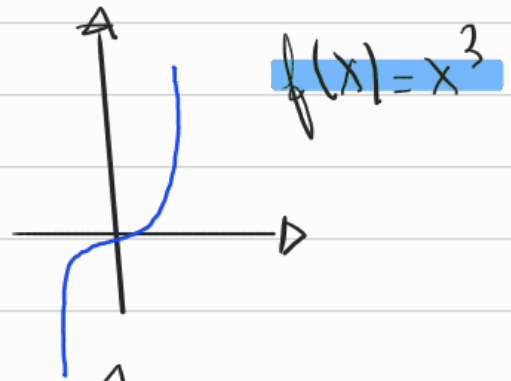


$$f(x) = -x^2$$

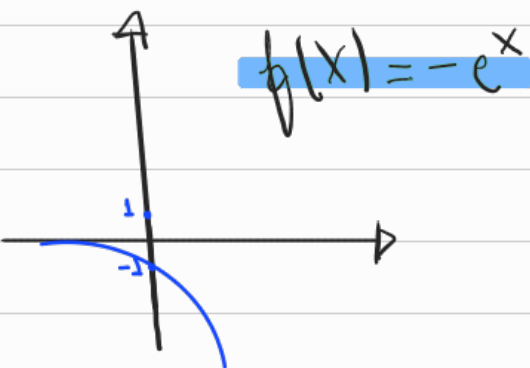
○ Modular:



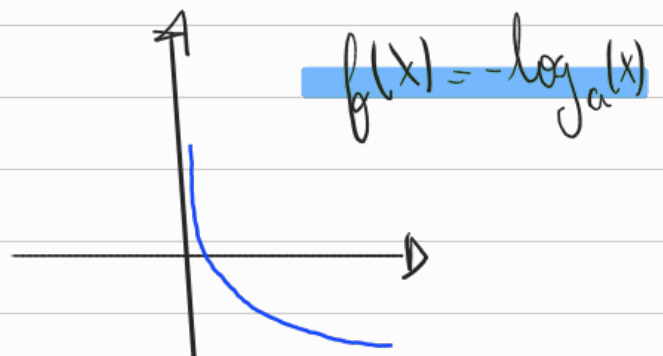
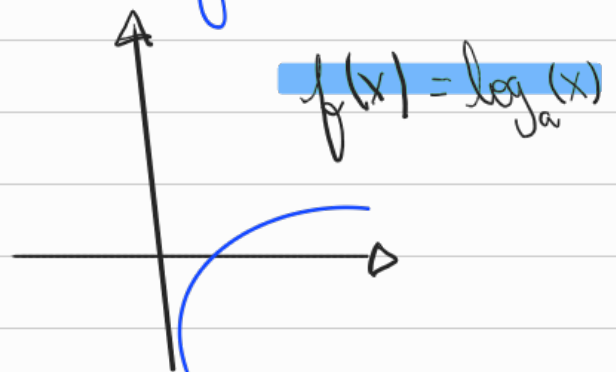
○ Polynomial °3:



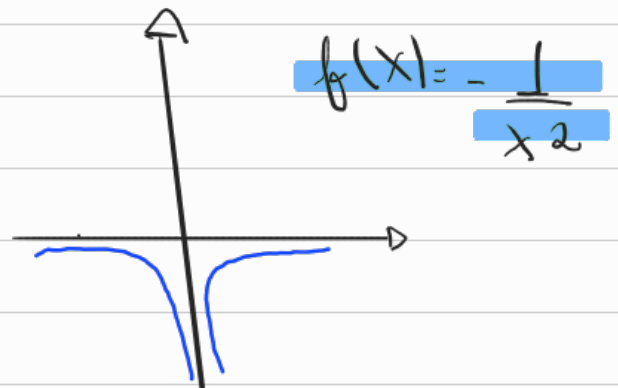
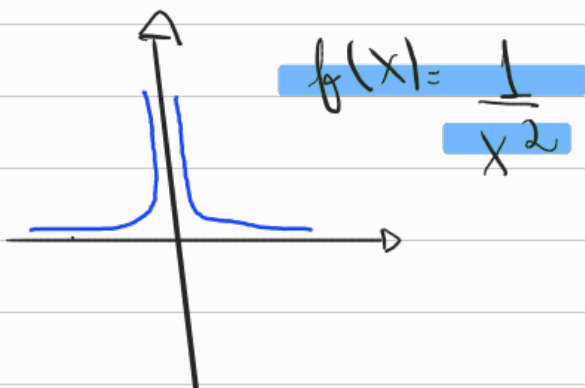
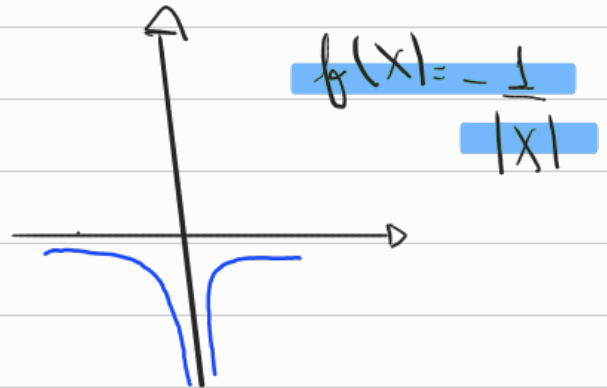
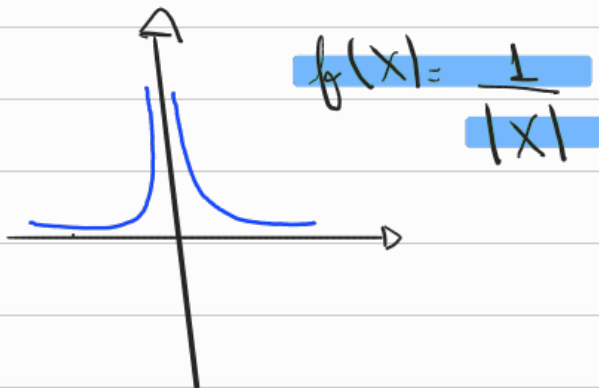
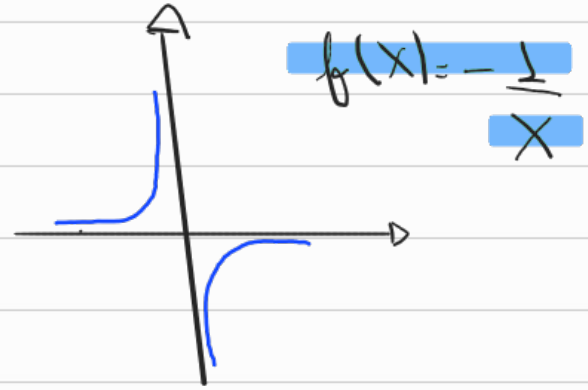
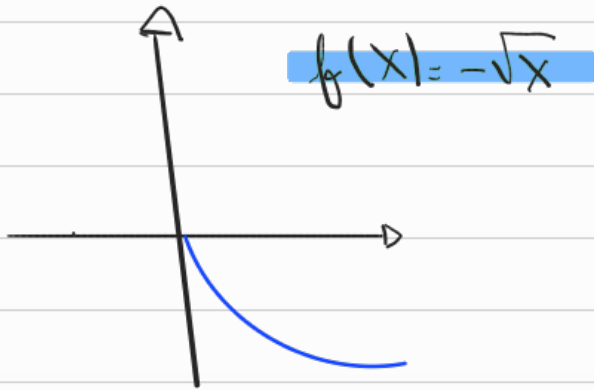
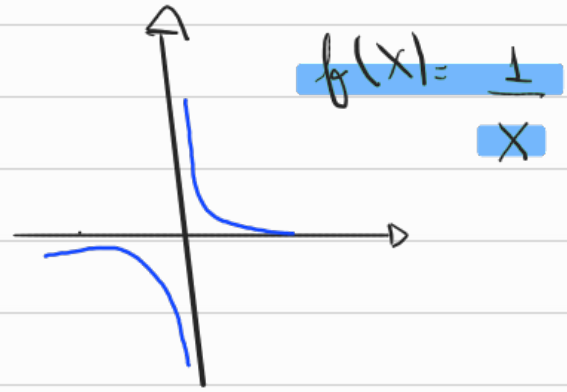
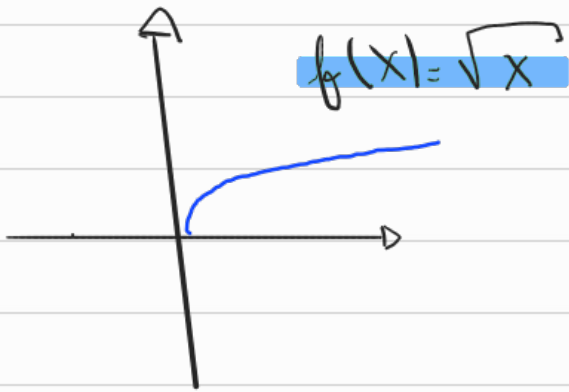
○ Exponencial



○ Logaritmica



o Outras:



Comportamento de Gráficos

Operações:

$$\begin{aligned} & f(x) + c \\ & f(x + c) \\ & -f(x) \\ & f(-x) \\ & a \cdot f(x) \\ & f(ax) \\ & f(x)^n \\ & \sqrt[n]{f(x)} \end{aligned}$$

Efeito:

Dobre ($c > 0$) ou abaixa ($c < 0$)
Esquerda ($c > 0$) ou direita ($c < 0$)
Reflete em relação ao x
Reflete em relação ao y
Estica ($a > 1$) ou comprime ($a < 1$)
Comprime ($a > 1$) ou estica ($0 < a < 1$)
 $n > 1$ $\rightarrow$ + inclinado
 $n > 1$ $\rightarrow$ menos inclinado

